Math 103 Class 11

1. Two events are **independent** if the fact that one occurs does not affect the probability of the other occurring.

   Examples:
   a.) flipping a coin and getting a head.
   flipping the same coin and getting a tail.
   b.) flipping a coin and getting a head.
   drawing a card and getting a queen.
   c.) drawing a card and getting an Ace of Clubs.
   replacing it . . .
   and drawing a card and getting a Queen of Diamonds.

   Examples:
   State which events are independent and which are dependent.
   (I) a.) Tossing a coin and drawing a card from a deck.
   (D) b.) Drawing a ball from an urn, not replacing it, and then drawing a second ball.
   (D) c.) Getting a raise in salary, and purchasing a new car.
   (D) d.) Driving on ice and having an accident.
   (I) e.) Having a large shoe size and having a high IQ.
   (D) f.) A father being left-handed and a daughter being left-handed.
   (D) g.) Smoking excessively and having lung cancer.
   (I) h.) Eating an excessive amount of ice cream and smoking an excessive amount of cigarettes.

2. **Multiplication Rule 1**

   When two events are independent, the probability of both occurring is: $P(A \text{ and } B) = P(A) \bullet P(B)$

   Example: What is the probability of tossing a head and following it up by tossing a tail?

   $$\begin{aligned} P(H \text{ and } T) &= P(H) \bullet P(T) \\ &= 1/2 \bullet 1/2 \\ &= 1/4 \end{aligned}$$

   Example: What is the probability of tossing a head and drawing the Ace of Hearts?

   $$\begin{aligned} P(H \text{ and } ACE) &= P(H) \bullet P(T) \\ &= 1/2 \bullet 1/52 = 1/104 \end{aligned}$$

Example: What is the probability of drawing the ACE of Hearts 2 times in a row, using replacement?

$$P(\text{Ace and Ace}) = P(\text{Ace}) \bullet P(\text{Ace})$$
$$= 1/52 \bullet 1/52$$
$$= 1/2704$$

Example: My chances of stumbling and falling on a flight of stairs are 0.04. If I use 4 flights of stairs today, what are my chances of falling on all four?

$(.04)^4 = .00000256$ Show how to use the "carat".

Example: Find the probability of picking 2 people at random who were born in the same month.

First person (can't be wrong)	12/12
Second person	1/12

$$12/12 \bullet 1/12 = 1/12$$

Find the probability of picking two people at random who were born in January.

$$1/12 \bullet 1/12 = 1/144$$

Example: What is the probability that a husband, wife and daughter have the same birthday?

$$365/365 \bullet 1/365 \bullet 1/365 = 1/133{,}225$$

3. **Dependent Events**

When the outcome or occurrence of the first event affects the outcome or occurrence of the second event in such a way that the probability is changed, the events are said to be **dependent**.

Examples:

a.) Drawing 4 Jacks in a row with no replacement.
b.) Drawing 2 diamonds in succession with no replacement.
c.) Three people in a row quickly go through a metal detector. It goes off. Which person triggered it? Search. . . The chances of finding the culprit increase with each person.

4. **Multiplication Rule 2**

When two events are dependent, the probability of both occurring is:

$$P(A \text{ and } B) = P(A) \bullet P(B|A)$$

Example: To get an A in the class.
10 slips of paper , numbered 1 - 10, are put in a hat. To get an A, you must get a 7, followed by a 6. What is the probability of winning?

$$\begin{aligned} P(7 \text{ and } 6) &= P(7) \bullet P(6|7) \\ &= 1/10 \bullet 1/9 \\ &= 1/90 \end{aligned}$$

Example: A box contains 5 white dice, 4 red dice, and 3 yellow dice. What is the probability of drawing a white, a red, and a yellow die in succession, with no replacement?

$$5/12 \bullet 4/11 \bullet 3/10 = 1/22$$

Example: In this class there are ____ guys and ____girls. If I select 2 of your homework papers at random, find the probability that both papers belong to girls.

$$28/39 \bullet 27/38 = 126/247 \quad \text{(example)}$$

Example: In a department store there are 120 customers, 90 of whom will buy at least one item. If 5 customers are selected at random, one by one, find the probability that all will buy at least one item.

$$\begin{aligned} 90/120 \bullet 89/119 \bullet 88/118 \bullet 87/117 \bullet 86/116 &= 42097/182546 \\ &= .2306 \end{aligned}$$

Example: In a scientific study there are 8 guinea pigs, 5 of whom are pregnant. If 3 are selected at random without replacement, find the probability that all are pregnant.

$$5/8 \bullet 4/7 \bullet 3/6 = 5/28$$

show calculator technique for working with fractions

5. **Conditional Probability**: The probability that event B occurs after event A has already occurred.

6. **Formula for Conditional Probability**:

The probability that the second event B occurs given that the first event A has occurred can be found by dividing the probability that both events occurred by the probability that the first event has occurred. The formula is:

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

Derivation of the formula:

Use Multiplication Rule 2:

$$\frac{P(A \text{ and } B)}{P(A)} = \frac{P(A) \bullet P(B|A)}{P(A)}$$

$$\frac{P(A \text{ and } B)}{P(A)} = P(B|A)$$

7. Example:

At a small college, the probability that a student takes physics and sociology is 0.092. The probability that a student takes sociology is 0.73. Find the probability that the student is taking physics, given that he/she is taking sociology.

$$P(\text{physics} \mid \text{sociology}) = \frac{P(\text{sociology and physics})}{P(\text{sociology})}$$

$$= .092 / .73$$

$$\approx .126 \approx .13$$

8. Example:

A circuit to run a model railroad has eight switches. Two are defective. If a person selects 2 switches at random and tests them, find the probability that the second one is defective, given that the first one is defective.

$$P(2^{nd} \mid 1^{st}) = \frac{P(1^{st} \text{ and } 2^{nd})}{P(1^{st})}$$

$$= \frac{(2/8)(1/7)}{2/8}$$

$$= 1/7$$

9. Example:

In a pizza restaurant, 95% of the customers' order pizza. If 65% of the customers' order pizza and a salad, find the probability that a customer who orders pizza will order a salad.

$$P(\text{salad} \mid \text{pizza}) = \frac{P(\text{pizza and salad})}{P(\text{pizza})}$$

$$= .65 \ / \ .95$$

$$= .684$$

10. Example:

Eighty students in a school cafeteria were asked if they favored a ban on smoking in the cafeteria. The results of the survey are shown in the table.

Class	Favor	Oppose	No Opinion	Totals
Frosh	15	27	8	50
Soph	23	5	2	30
Totals	38	32	10	80

P(Favor) = 38/80 = 19/40
P(Oppose) = 32/80 = 2/5

$P(\text{Frosh}) = 50/80 = 5/8$

$P(\text{Soph}) = 30/80 = 3/8$

$$P(\text{Favor or Frosh}) = P(\text{Favor}) + P(\text{Frosh}) - P(\text{Favor and Frosh}) = 38/80 + 50/80 - 15/80 = 73/80$$

$P(\text{Oppose and Soph}) = 5/80 = 1/16$

$$P(\text{Oppose} \mid \text{Frosh}) = \frac{P(\text{O and F})}{P(\text{F})} = \frac{(27/80)}{(50/80)} = 27/50$$

$$P(\text{Soph} \mid \text{Favor}) = \frac{P(\text{S and F})}{P(\text{Favor})} = \frac{(23/80)}{(38/80)} = 23/38$$

$$P(\text{N.O.} \mid \text{Frosh}) = \frac{P(\text{N.O. and Frosh})}{P(\text{Frosh})} = \frac{(8/80)}{(50/80)} = 8/50 = 4/25$$

11. Example of “at least”

There are five chemistry instructors and six physics instructors at a college. If a committee of four instructors is selected, find the probability of at least one of them being a physics instructor.
One option is to find each separately. (PI = physics instructor)
P(1 PI) + P(2 PI) + P(3 PI) + P(4 PI)

But, it is easier to do this:

1 - P(no PI)
1 - [(5/11)(4/10)(3/9)(2/8)]
1 - 1/66
65/66

12. A “lot” of portable radios contain 15 good and 3 defective ones. If 2 are selected and tested, find the probability that at least one will be defective.

1 - P(no defectives)
1 - [(15/18)(14/17)]
1 - (35/51)
16/51

Alternate Method:

13. Example:

A medication is 75% effective against a bacterial infection. Find the probability that if 12 people take the medication, at least one person's infection will not improve.

$$1 - P(\text{will improve})$$
$$1 - (.75)^{12}$$
$$1 - 0.0317$$
$$0.9683$$

Alternate Method: (much too long)

14. Sample space for tossing dice:

Roll a die 1, 2, 3, 4, 5, 6

Roll 2 dice

(1,1),(1,2),(1,3),(1,4),(1,5),(1,6)
(2,1),(2,2),(2,3),(2,4),(2,5),(2,6)
(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)
(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)
(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)
(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)